



ID number _____ Date _____ Group _____

Signature

- 1) Which of the following options evaluates correctly $(f \circ g)(x)$

a) $g(f(x))$ b) $g(x)f(x)$ c) $f(g(x))$ d) $f(x)g(x)$
- 2) Which of the following functions models exponential decay.

a) $m(t) = Ae^{-r+t}$ b) $m(t) = -Ae^{r-t}$ c) $m(t) = Ae^{-rt}$ d) $m(t) = -Ae^{-rt}$
- 3) Considering $f(x) = (x + 1)^2 - 1$. What would the new function be like after shifting 5 units to the left and 3 units down?

a) $f(x) = (x + 5)^2 - 3$ b) $f(x) = (x + 2)^2 - 4$ c) $f(x) = (x + 2)^2 + 2$ d) $f(x) = (x + 6)^2 - 4$
- 4) Given the function $f(x) = -2\sqrt{x^2 - 4} - 5$, which of the following represents its domain?

a) $-\infty < x < \infty$ b) $-\infty < x \leq 4$ c) $-\infty < x \leq 0$ d) $2 \leq x < \infty$

- 1) Solve the following equations. Simplify the answer as much as possible and check that all solutions satisfies the original equation.

a. $\log(3-x) + \log(x+9) = 0$

b. $6^{x-3} - 2^{x+2} = 0$

PrepaTec Eugenio Garza Lagüera
Department of Mathematics
Algebraic and Transcendental Functions

- 2) For the functions $f(x) = \frac{5x}{x+2}$ and $g(x) = \frac{8}{x}$ find the indicated composition and reduce the expression as much as possible,

a. $(f \circ g)(x)$	b. $(g + g)(x)$

- 3) Use logarithmic properties to condense the following expression $2 \log_8 y + \frac{1}{3}(\log_8 x - 2 \log_8 w)$

Section 3: Function analysis

- 1) Find the inverse function of $f(x) = \frac{2x-4}{3x-10}$. Reduce the expression as much as possible and proof your answer by evaluating the function composition $(f \circ f^{-1})(x) = x$.

PrepaTec Eugenio Garza Lagüera
Department of Mathematics
Algebraic and Transcendental Functions

- 2) Find the roots of the function $f(x) = \frac{x^2+x-2}{x+2}$.

Modeling with math

- 1) Hydrogen potential (pH) is a number used to describe the acidity or basicity of a chemical substance and is defined by the equation $pH = -\log(H^+)$, where H^+ measures the concentration of hydrogen ions moles per liter.
 1. Determine the pH of a coke if $H^+ = 1 \times 10^{-3}$.
 2. What is the hydrogen ions concentration of the stomach acid with pH of 2?

PrepaTec Eugenio Garza Lagüera
Department of Mathematics
Algebraic and Transcendental Functions

- 2) The half-life of Radium-266 is 1600 years. Suppose that there is a sample of 22mg. Recalling that the mass decays exponentially, fulfill the following points
- i) Write the function that models the remaining mass after t years.
 - ii) How much mass will remain after 4000 years?
 - iii) How much time will it take for the sample to be only a quarter of its initial value?.

Conic Sections

- 1) Identify the conic section given the following general equation $16x^2 - 9y^2 - 96x - 54y - 81 = 0$.